

Ankit Kumar

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Final-year CS student and Codeforces Expert specializing in backend architecture. Passionate about applying algorithmic problem-solving to build high-throughput APIs and concurrent, low-level infrastructure.

TECHNICAL SKILLS

Languages Java, C++ , SQL

Backend Spring Boot, Java NIO, Virtual Threads (Java 21), REST APIs

Systems Concurrency, Event-driven I/O, WebSocket, WebRTC, Replication Protocols

Databases PostgreSQL, Redis

Infra Prometheus, Grafana, AWS, Oracle Cloud Infrastructure, Git

ACHIEVEMENTS

- **TCS CodeVita Season 13** — **Global Rank 15** out of 669,000+ participants
- **Codeforces Expert** (rating 1655) · **LeetCode** — active problem solver across DSA domains

WORK EXPERIENCE

Software Engineer Intern

Feb 2025 – May 2025

ControlShift Talent Pvt. Ltd.

- Built WebRTC signaling server and WebSocket layer with heartbeat, room management, and reconnection for live candidate-AI video sessions
- Offloaded blocking email I/O via async job queues with retry and dead-letter handling; developed 30+ REST endpoints with centralized error handling

PROJECTS

StrataDB — Embedded Key-Value Storage Engine

github.com/ankitkr-04/strata-db

C++ 26, CMake, NUMA, Lock-Free Concurrency

- Engineered a NUMA-aware Arena allocator with mmap-backed blocks and opportunistic huge-page fallback, paired with thread-local allocation buffers (TLAB) to completely bypass global CAS contention
- Delivered **~48.8M ops/sec** at 4 threads ($1.47\times$ faster than malloc), slashing p99 tail latency from 577 ns to **53 ns** by reducing hot-path atomic operations from 2.86 to near-zero
- Built a lock-free skip-list MemTable with CAS-based insertion, append-versioned key ordering (key asc, seq desc), and flush/stall backpressure; validated under TSAN with 16-thread oversubscription stress tests
- Implemented epoch-based memory reclamation with per-thread slab storage and adaptive backoff, enabling safe deferred pointer reclamation across concurrent readers and writers
- Designed WAL staging path with thread-local block assembly, tearing-matrix block headers, and mutex-protected ready-queue handoff for flush pipeline integration

TicketLedger — High-Concurrency Cinema Booking Platform

github.com/ankitkr-04/ticket-ledger

Java 21, Spring Boot, PostgreSQL, Stripe

- Enforced zero double-bookings via PostgreSQL SELECT FOR UPDATE with strict lock ordering; analytically bounded hot-seat throughput at **~200 ops/sec** per row from lock-commit latency
- Scaled API layer with Java 21 Virtual Threads: 1M parked connections per JVM at **~1 KB/thread**, isolating HikariCP (20 connections) as the sole bottleneck
- Guaranteed exactly-once payment execution via idempotent Stripe webhook processing with strict financial consistency over UX in late-payment edge cases

Axiom — Redis-Compatible In-Memory Store

github.com/ankitkr-04/axiom

Java, NIO, Concurrency

- Delivered **~81K GET ops/sec** and **~68K SET ops/sec** (50 concurrent clients; $p99 \leq 3.1$ ms) via a single-threaded NIO event loop with direct ByteBuffer access, eliminating thread contention and GC pressure
- Implemented 29+ Redis-compatible commands: Strings, Lists, Sorted Sets (skip-list), Geospatial (geohash), Streams, and Pub/Sub with pattern matching
- Replicated Redis PSYNC protocol: RDB snapshot on initial sync, async incremental streaming, and WAIT-based sync guarantees; tracked per-replica lag via REPLCONF GETACK

EDUCATION

B.Tech in Computer Science

Aug 2023 – May 2026

Lakshmi Narain College of Technology, Bhopal

CGPA: **8.41 / 10**

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Developer Professional (1Z0-1084-25) — Oracle